What do we know about climate change?

Raffaele Ferrari, MIT

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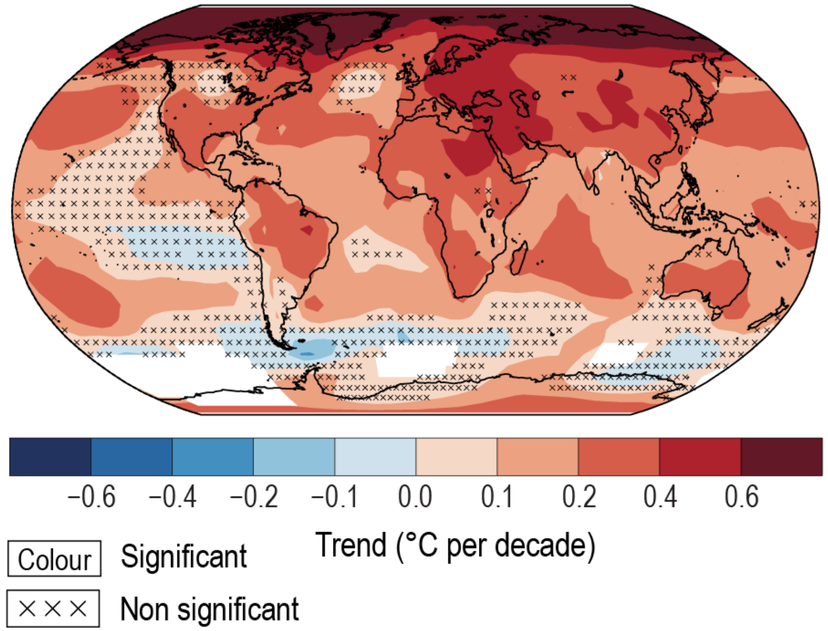
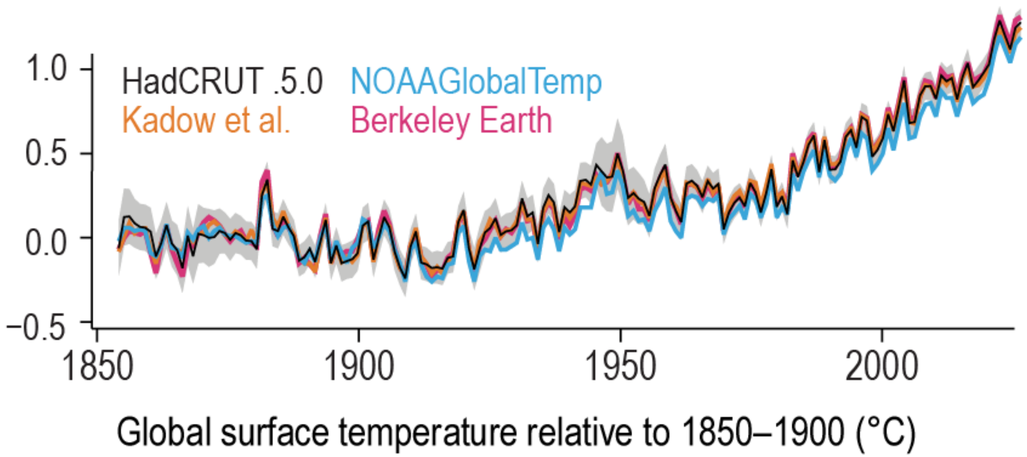
The Earth is warming: why?

Mean temperature change between 1981 and 2020



Global average temperature change from 1850



IPCC Sixth Assessment Report, 2023

Key points about climate science

- Earth’s climate is stable within certain limits, but sensitive to change in forcing within those limits
- Climate science has a long and illustrious history
- The idea that we are altering climate is based on simple physics, simple models, and complex models
- Human activities can and do have a strong effect on climate

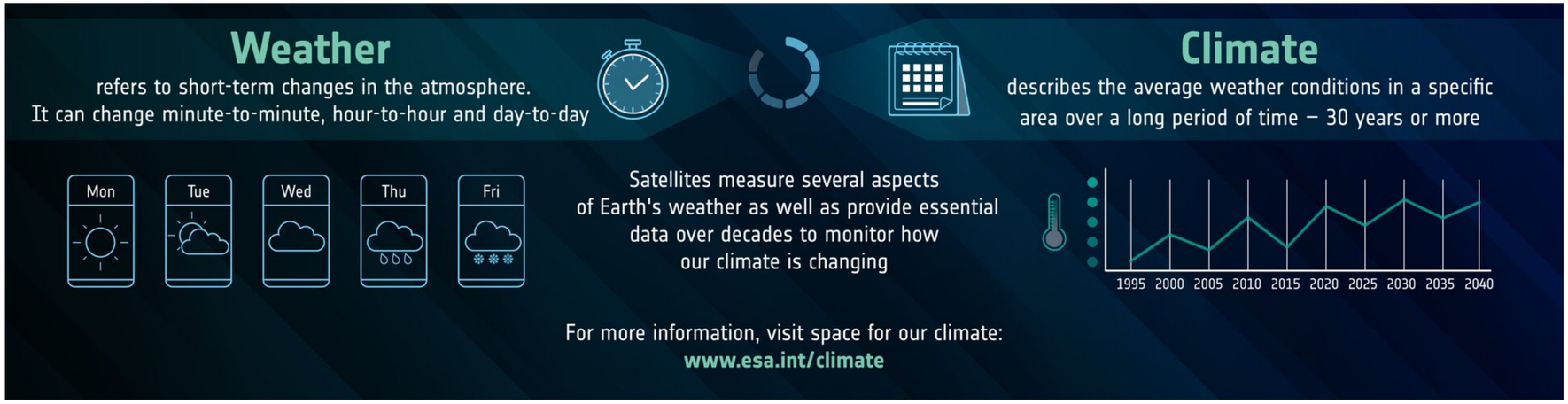
Definition of climate

Popular definitions:

- Climate is the average weather
- Climate is what you expect, weather is what you get

Technical usage:

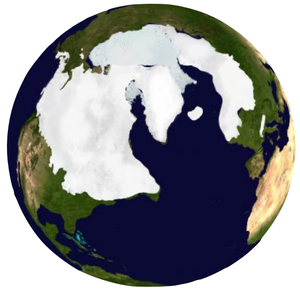
- “Climate” is the *statistics* of weather
- This includes not just the average weather, but the statistics of its variability
- Commonly calculated over periods of one decade or more

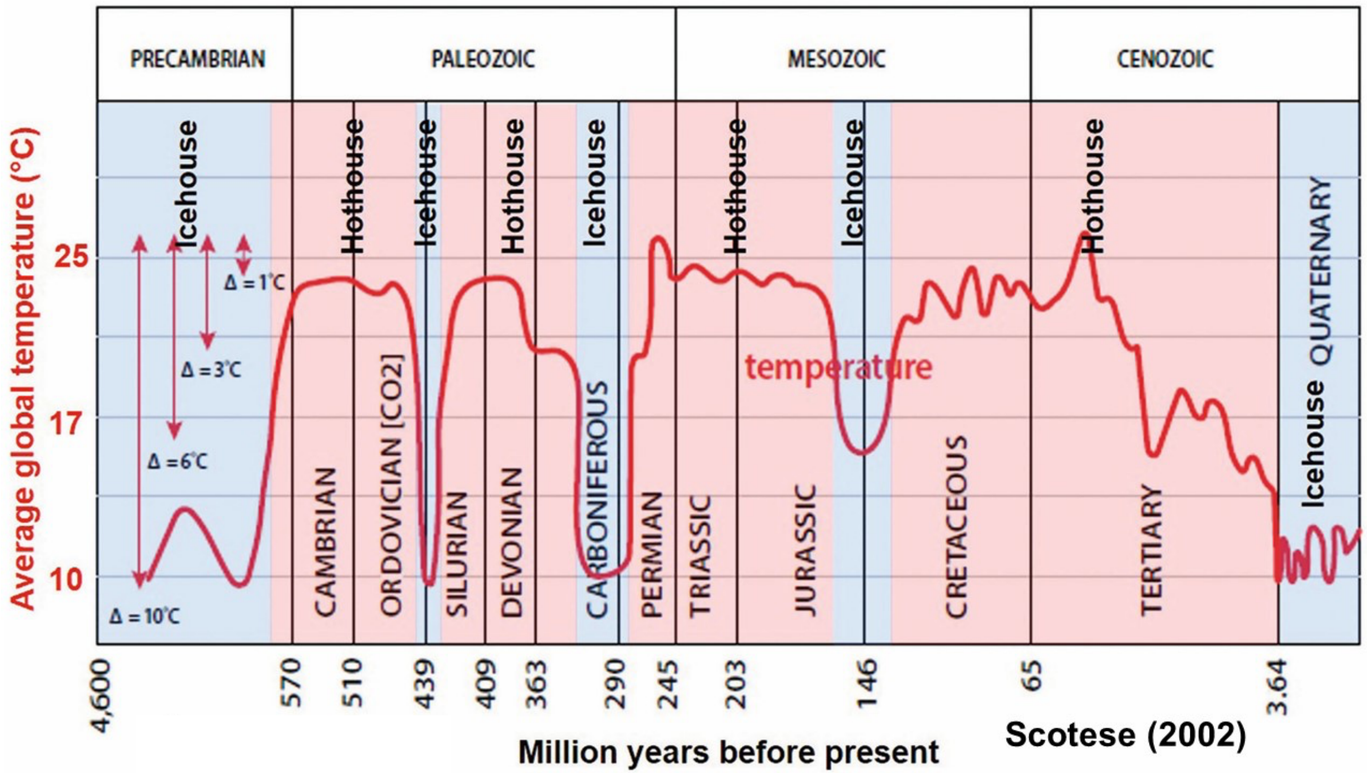


Climate change through geologic time

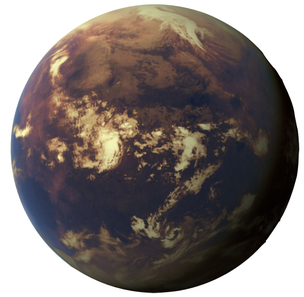
History of Earth's climate

Icehouse climates





Hothouse climates



Eocene Arctic Vegetation

Unicellular organisms

Plants, fungi, invertebrates, fish

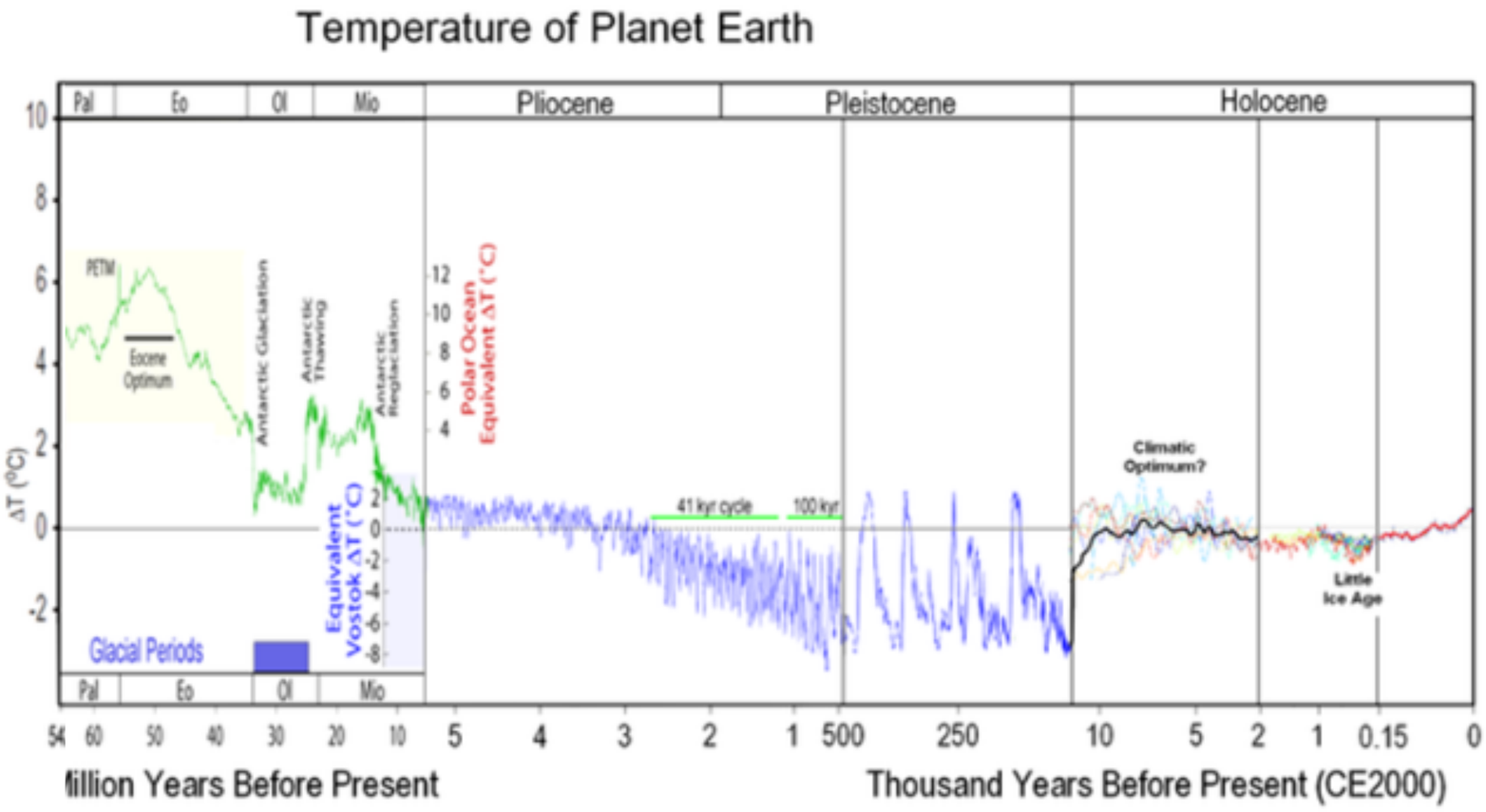
Ferns, reptiles, dinosaurs

Mammals

Primates

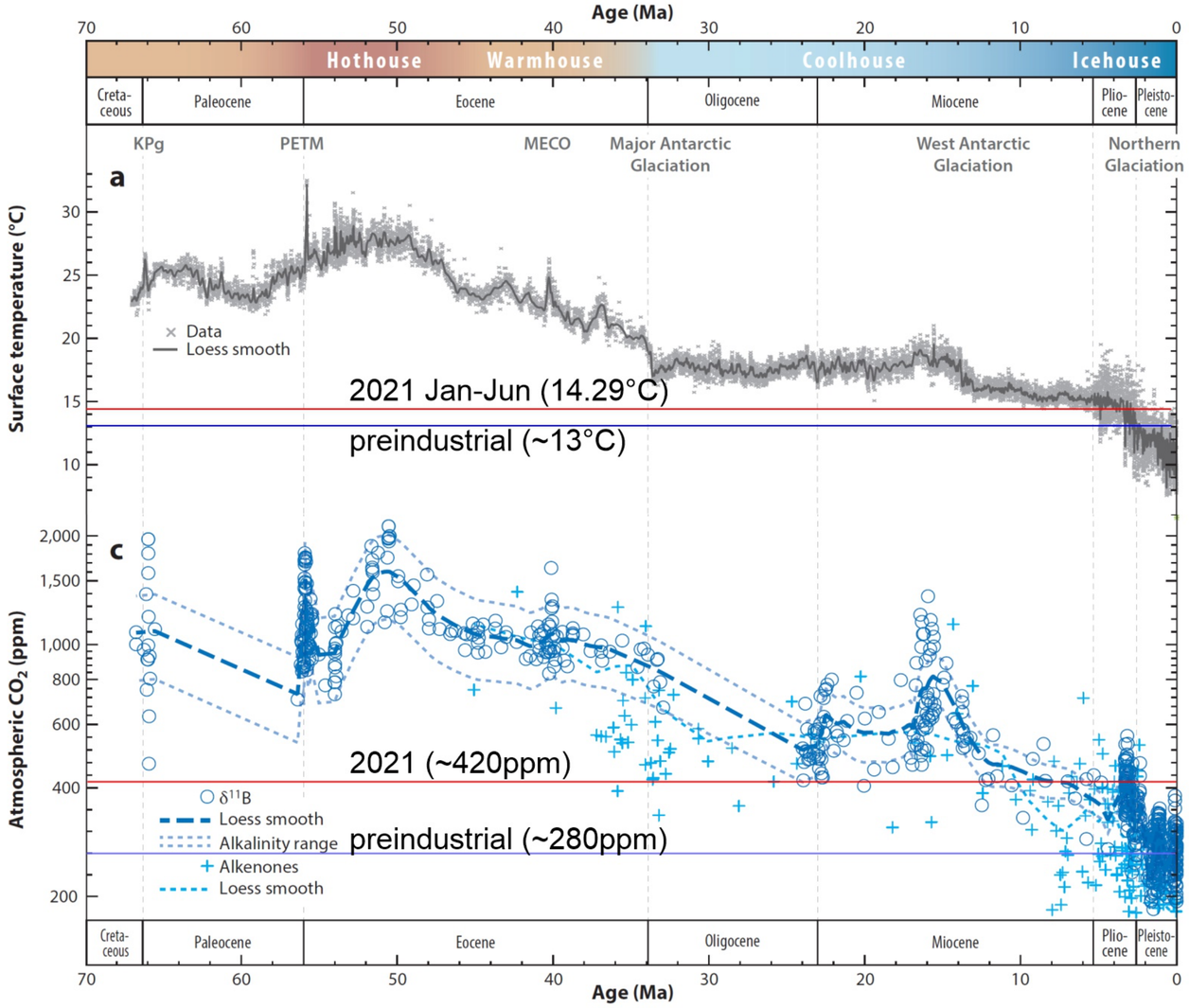
Cenozoic climate

65 million years of climate change



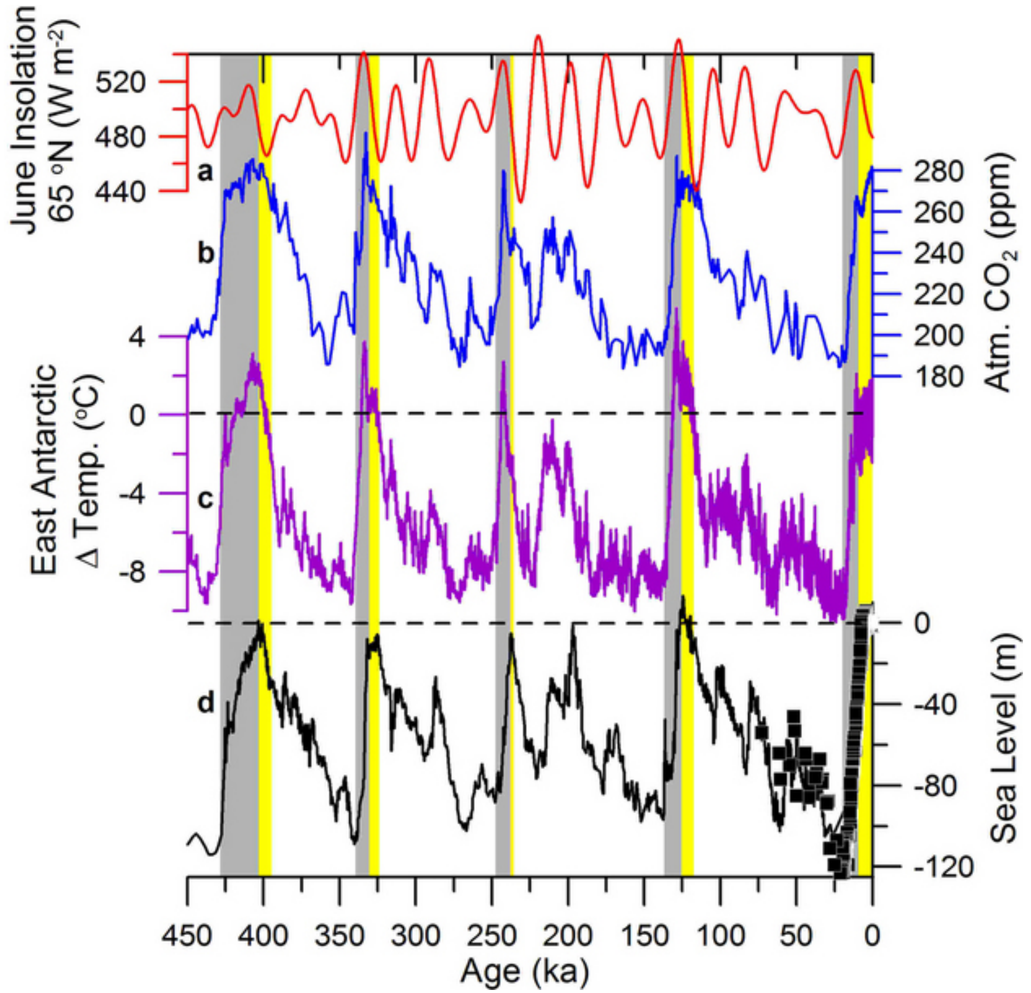
Wikipedia compilation

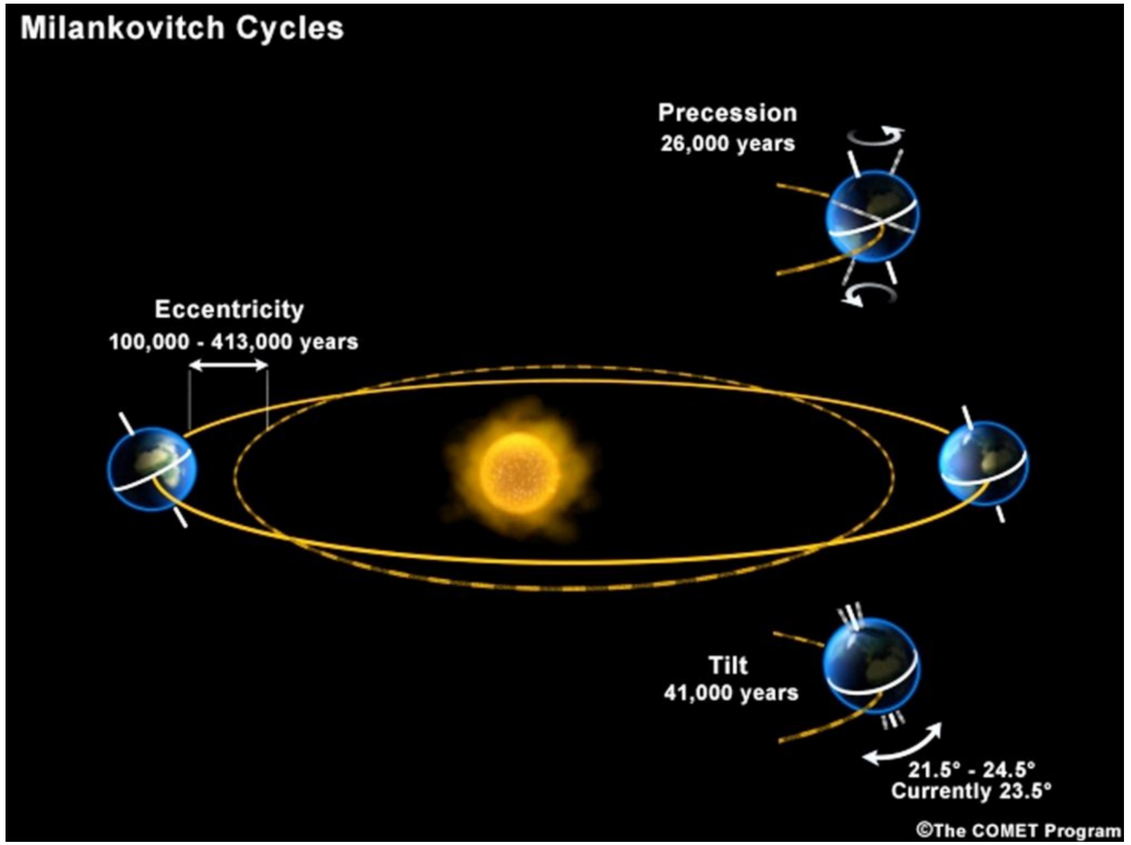
Cenozoic climate and CO_2



Rae et al. 2021

Quaternary ice ages, orbital forcing and CO_2

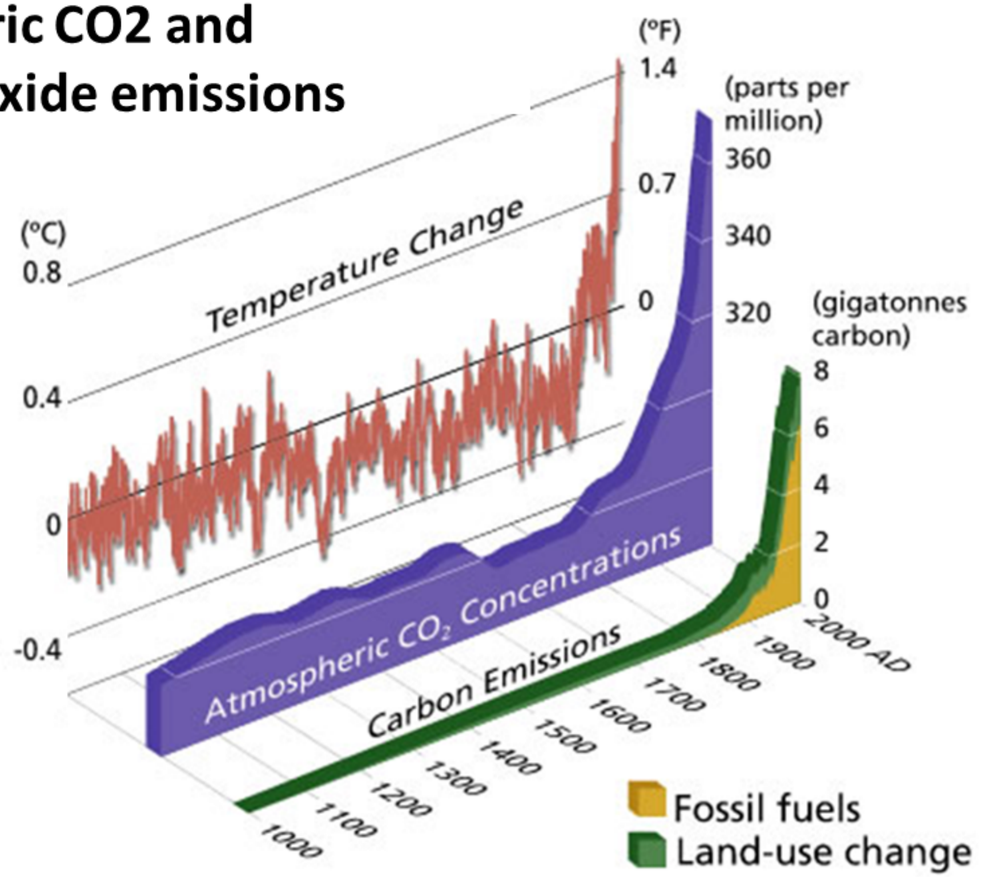




Carlson, 2011

The last 1000 years

Past 1000 years of temperature change, atmospheric CO2 and carbon dioxide emissions



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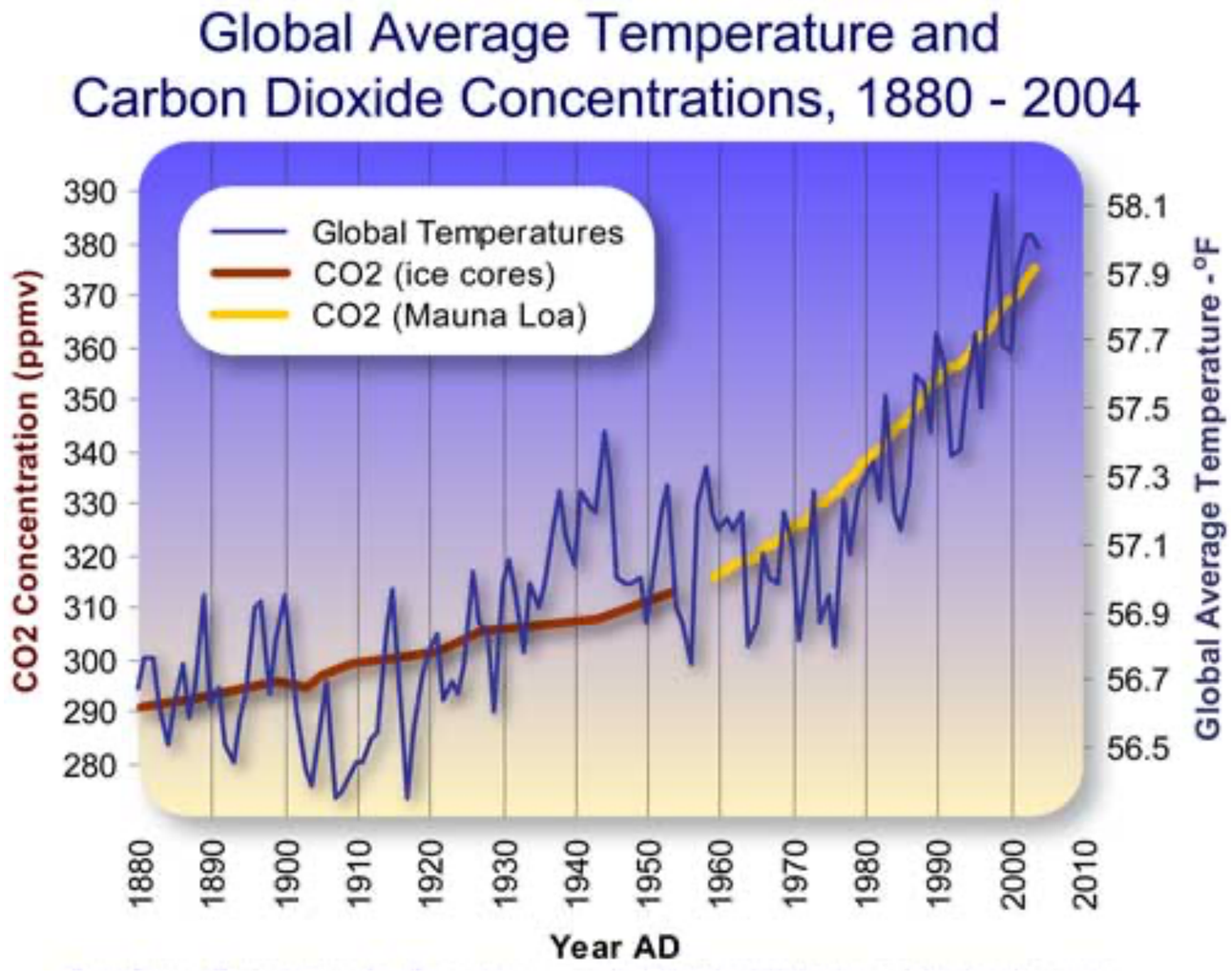
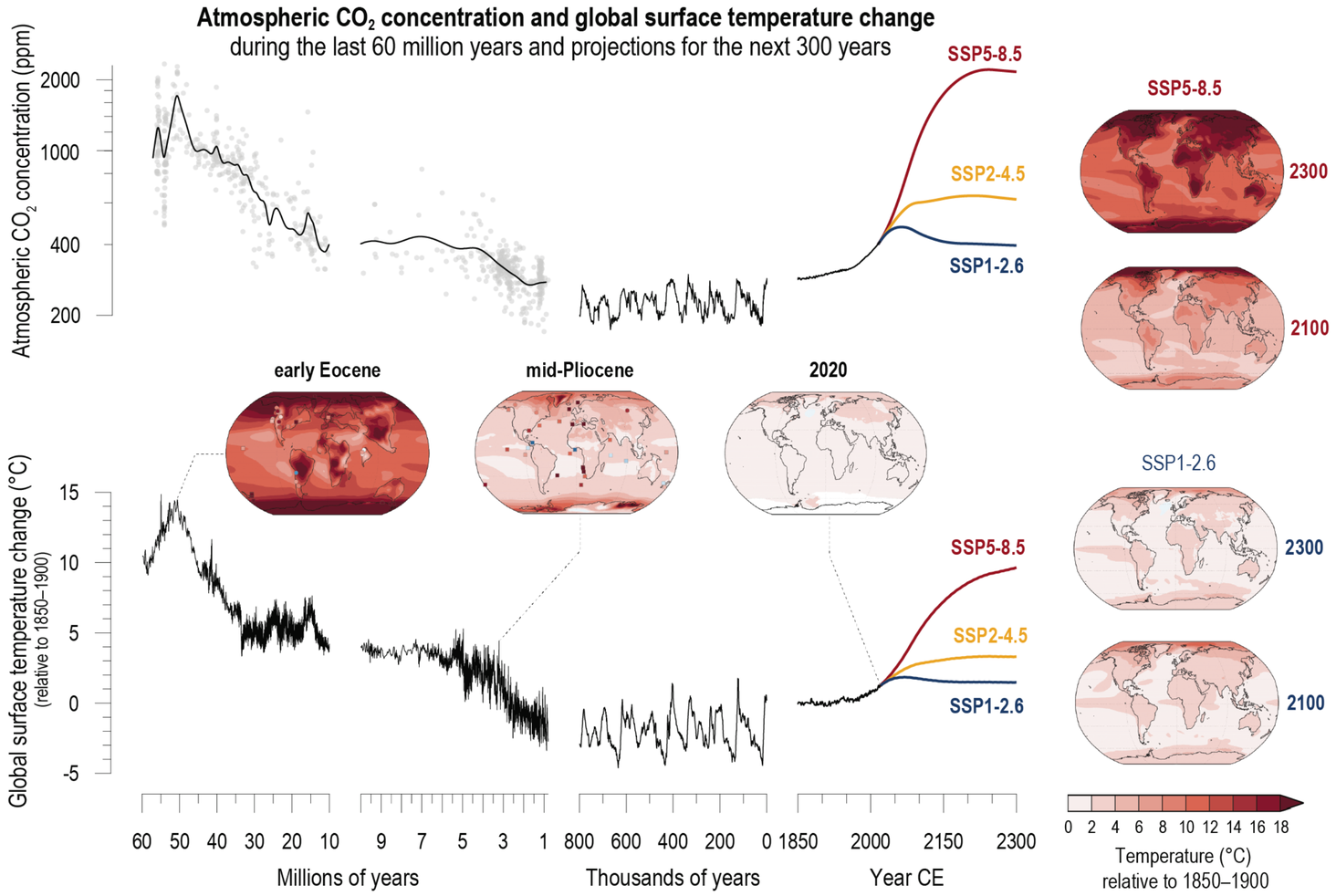
The last century



Michael Ernst, The Woods Hole Research Center Report

Future climate and CO_2



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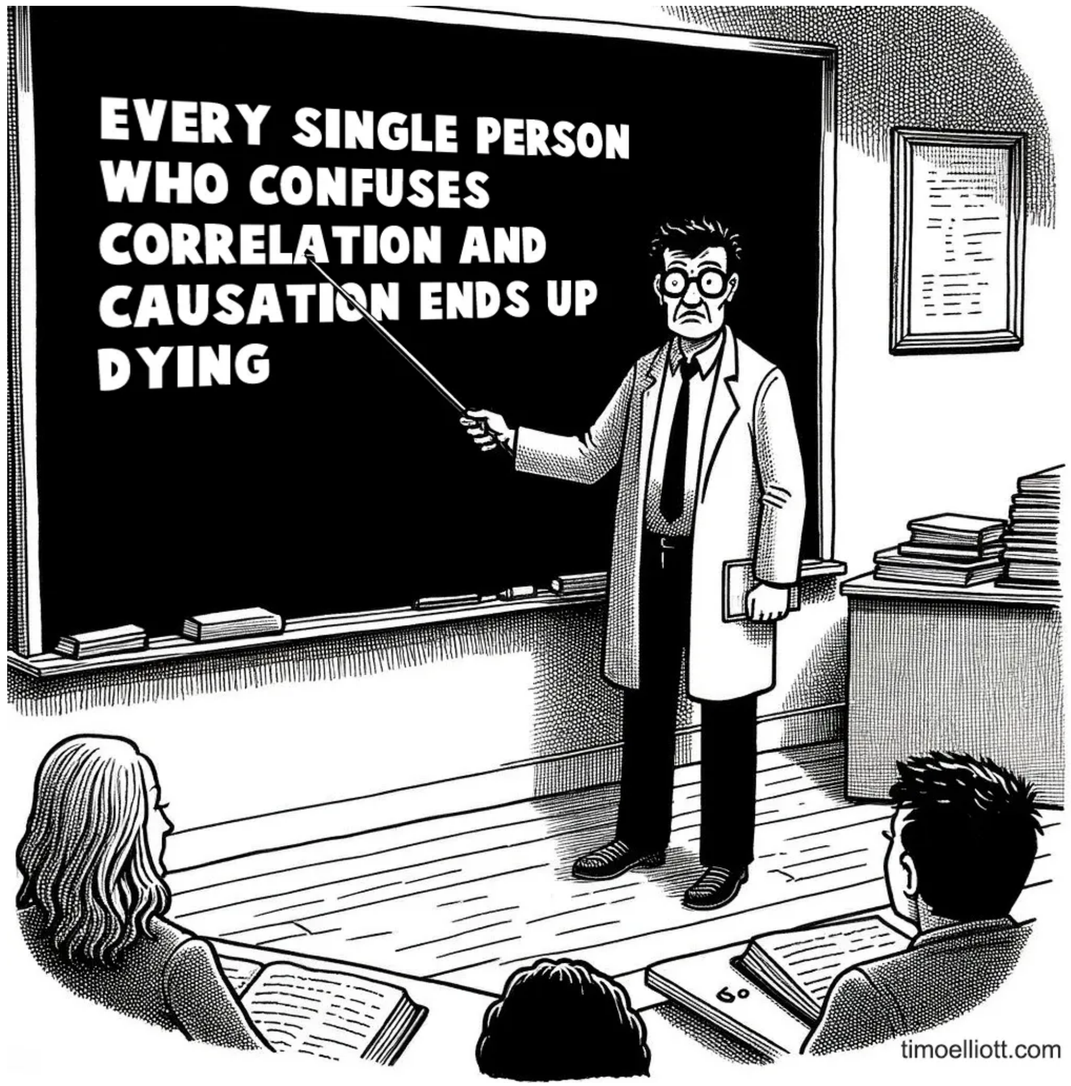
Correlation does not imply causation



Hannah Fry and timeelliot.com

Climate science

The greenhouse effect

THE

AMERICAN

JOURNAL OF SCIENCE, &c.

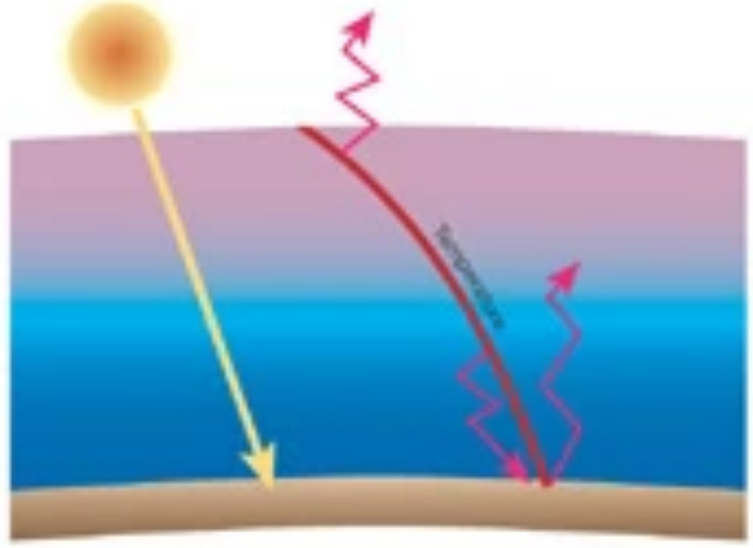
ART. I.—*General Remarks on the Temperature of the Terrestrial Globe and the Planetary Spaces*; by Baron FOURIER.*

Translated from the French, by Mr. EBENEZER BURGESS, of Amherst College.

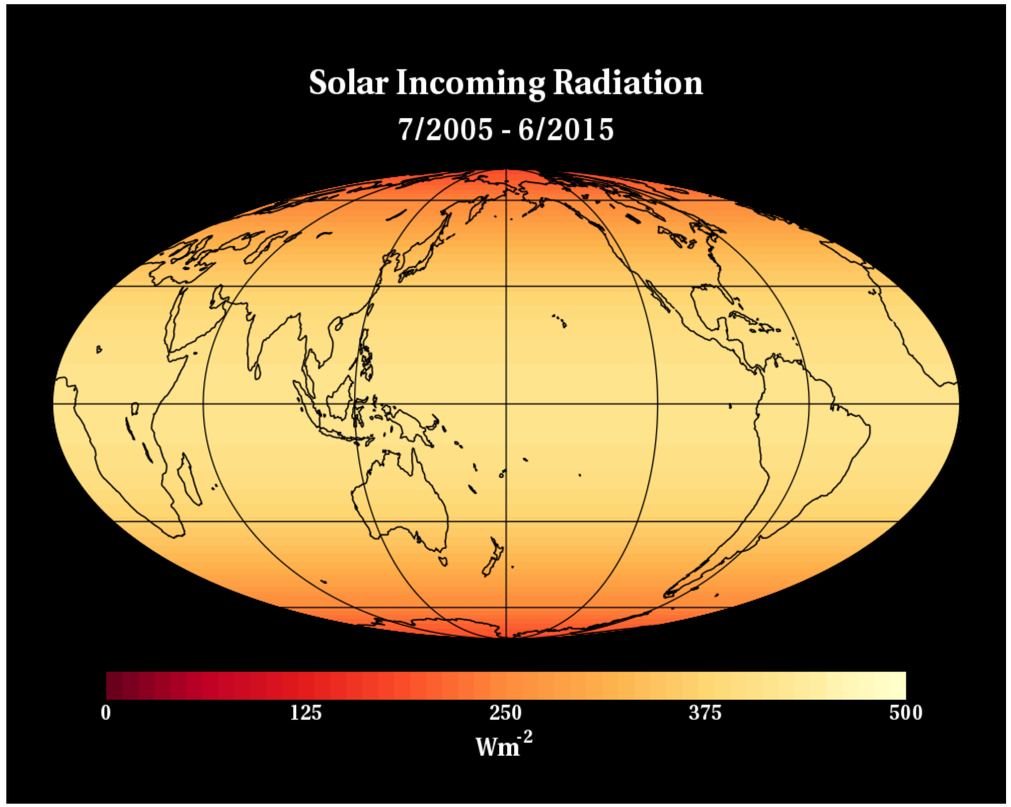
THE question of terrestrial temperature, one of the most remarkable and difficult in natural philosophy, involves very different elements which require to be considered in a general light. I have thought it would be useful to have condensed in a single essay, all the results of this theory. The analytical details here admitted, are found in works which I have already published. I was specially desirous of presenting to philosophers, in a concise table, a complete view of the phenomena and the mathematical relations which exist between them.

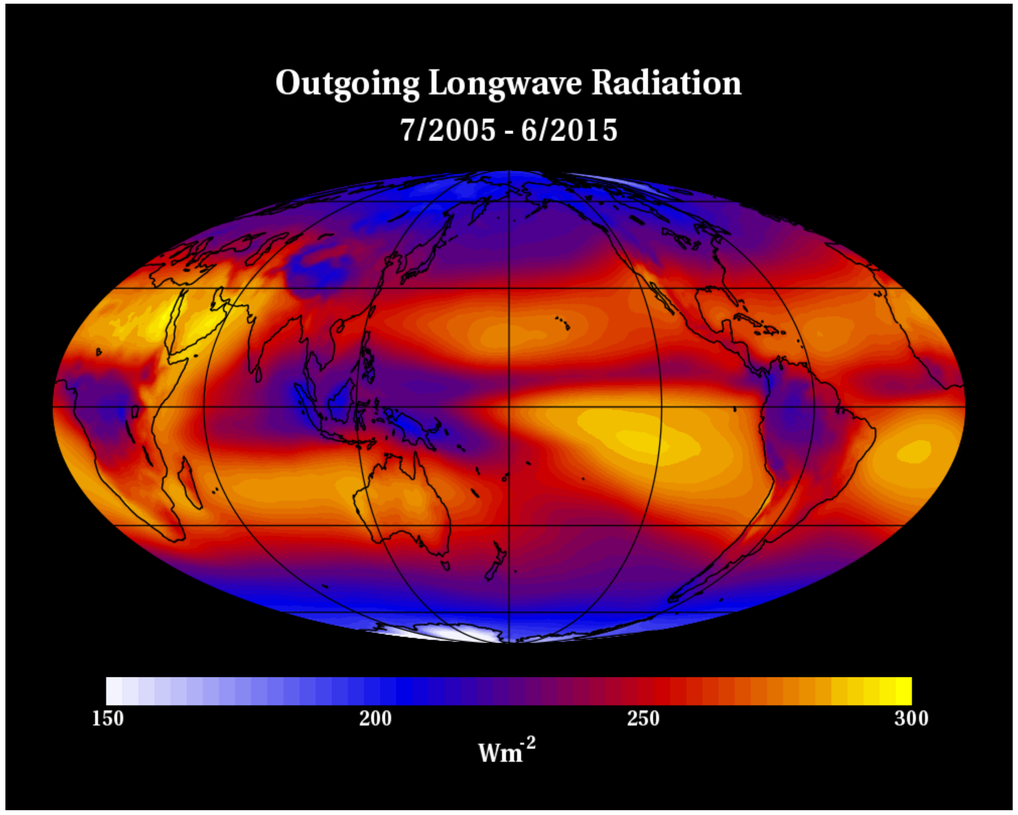
Jean Baptiste Fourier

(1768-1830)



The greenhouse effect





Incoming shortwave radiation and outgoing long wave radiation as measured by the CERES NASA satellite

The greenhouse effect

Eunice Foote

(1819-1888)

382 *On the Heat in the Sun's Rays.*

ART. XXXI.—*Circumstances affecting the Heat of the Sun's Rays;* by EUNICE FOOTE.

(Read before the American Association, August 23d, 1856.)

MY investigations have had for their object to determine the different circumstances that affect the thermal action of the rays of light that proceed from the sun.

Several results have been obtained.

First. The action increases with the density of the air, and is diminished as it becomes more rarified.

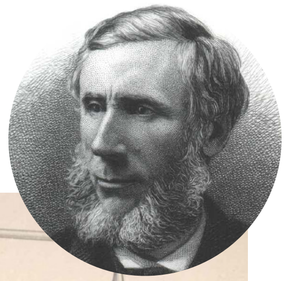
The experiments were made with an air-pump and two cylindrical receivers of the same size, about four inches in diameter and thirty in length. In each were placed two thermometers, and the air was exhausted from one and condensed in the other. After both had acquired the same temperature they were placed in the sun, side by side, and while the action of the sun's rays rose to 110° in the condensed tube, it attained only 88° in the other. I had no means at hand of measuring the degree of condensation or rarefaction.

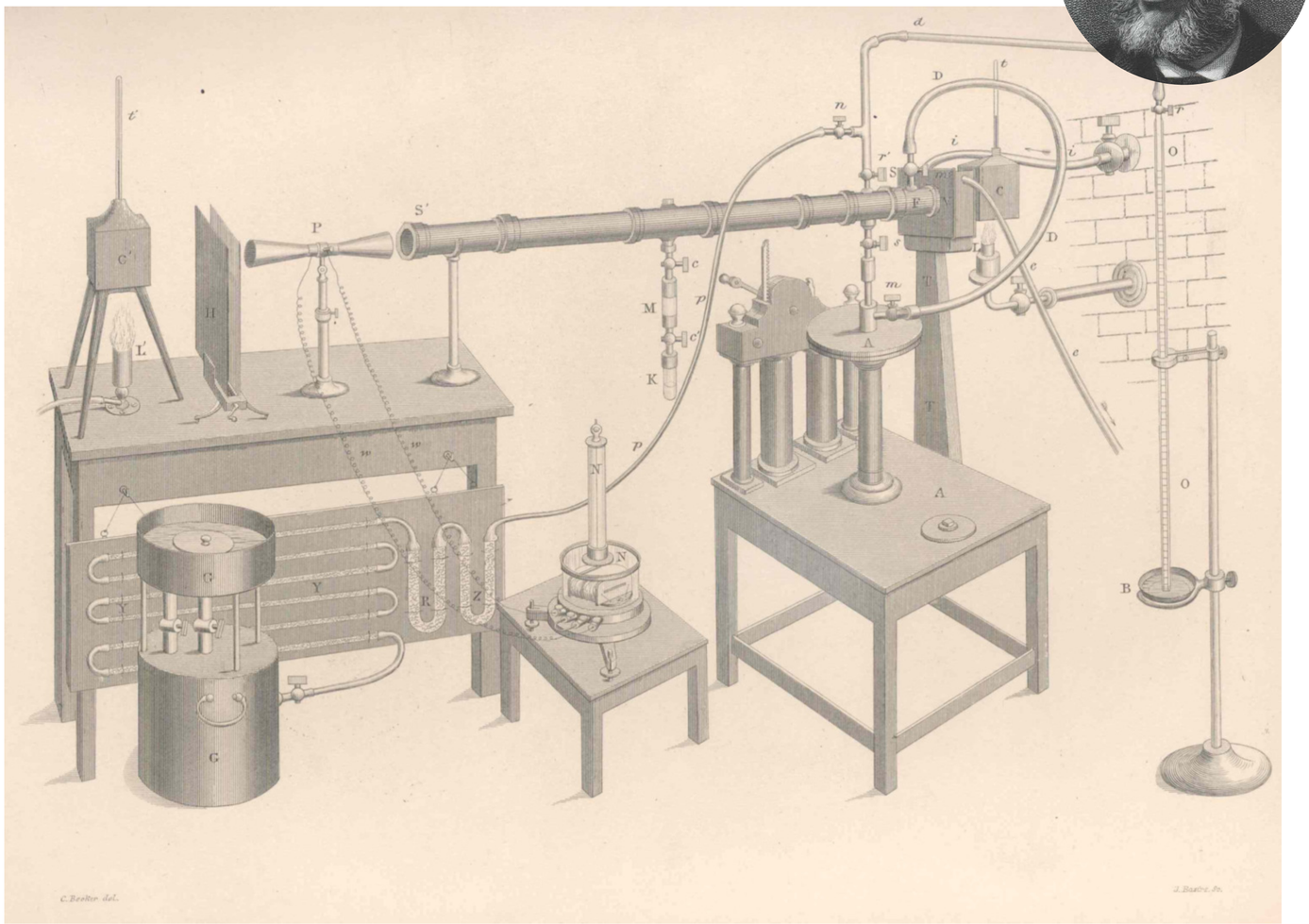
The observations taken once in two or three minutes, were as follows:

Exhausted Tube		Condensed Tube.	
In shade.	In sun.	In shade.	In sun.
75	80	75	80
76	82	78	95
80	82	80	100
83	86	82	105
84	88	85	110

John Tyndall

(1822-1893)

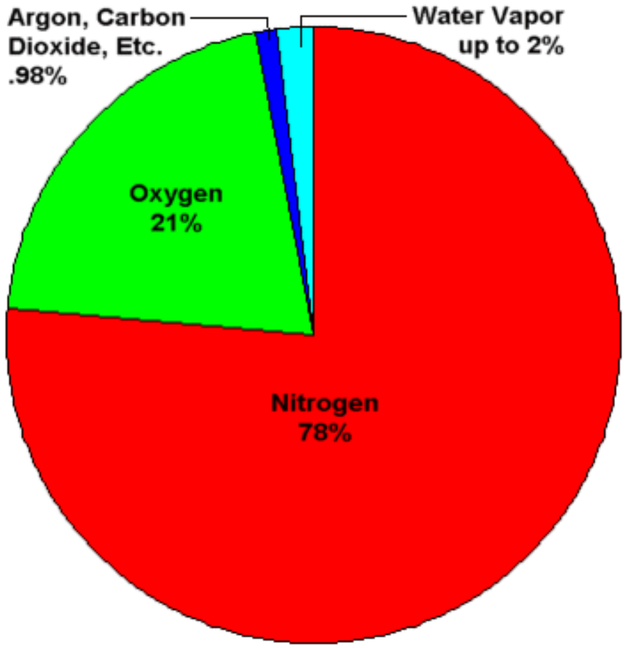




The greenhouse effect

- Oxygen (O_2), nitrogen (N_2), and argon (Ar), though they make up ~99% of the atmosphere, are almost entirely transparent to solar and terrestrial radiation

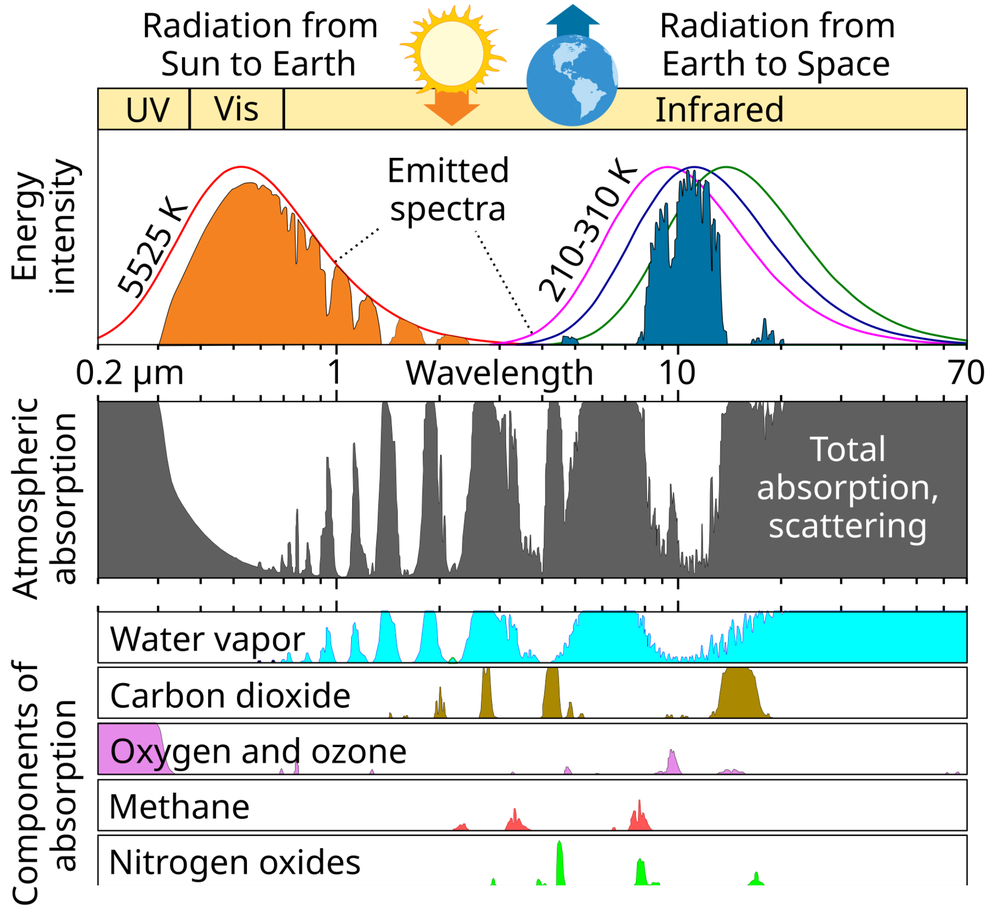
- Water vapor (H_2O), carbon dioxide (CO_2), nitrous oxide (N_2O), and a handful of other trace gases make the lower atmosphere nearly opaque to infrared radiation, though still largely transparent to solar radiation (but clouds have strong effects on radiation at all wavelengths). Together they increase the Earth's surface temperature from about -18°C to around 15°C



The greenhouse effect

- Water Vapor (H_2O), about 0.25% of the mass of the atmosphere, is the most important greenhouse gas, but responds to atmospheric temperature change on a time scale of about 2 weeks

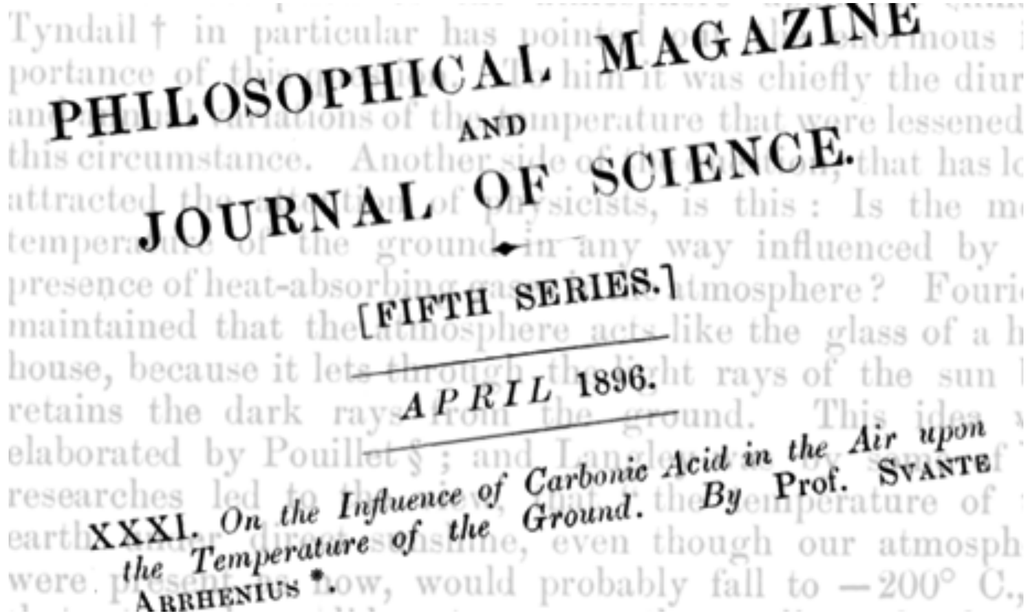
- Climate is strongly influenced by long-lived greenhouse gases (e.g. CO_2, CH_4, N_2O) that comprise about 0.04% of the mass of the atmosphere. Concentration of CO_2 has increased by 50% since the dawn of the industrial revolution



The greenhouse effect

“Any doubling of the percentage of carbon dioxide in the air would raise the temperature of the earth's surface by 4°; and if the carbon dioxide were increased fourfold, the temperature would rise by 8°.”

Världarnas utveckling (Worlds in the Making), 1906



PHILOSOPHICAL MAGAZINE
AND
JOURNAL OF SCIENCE.

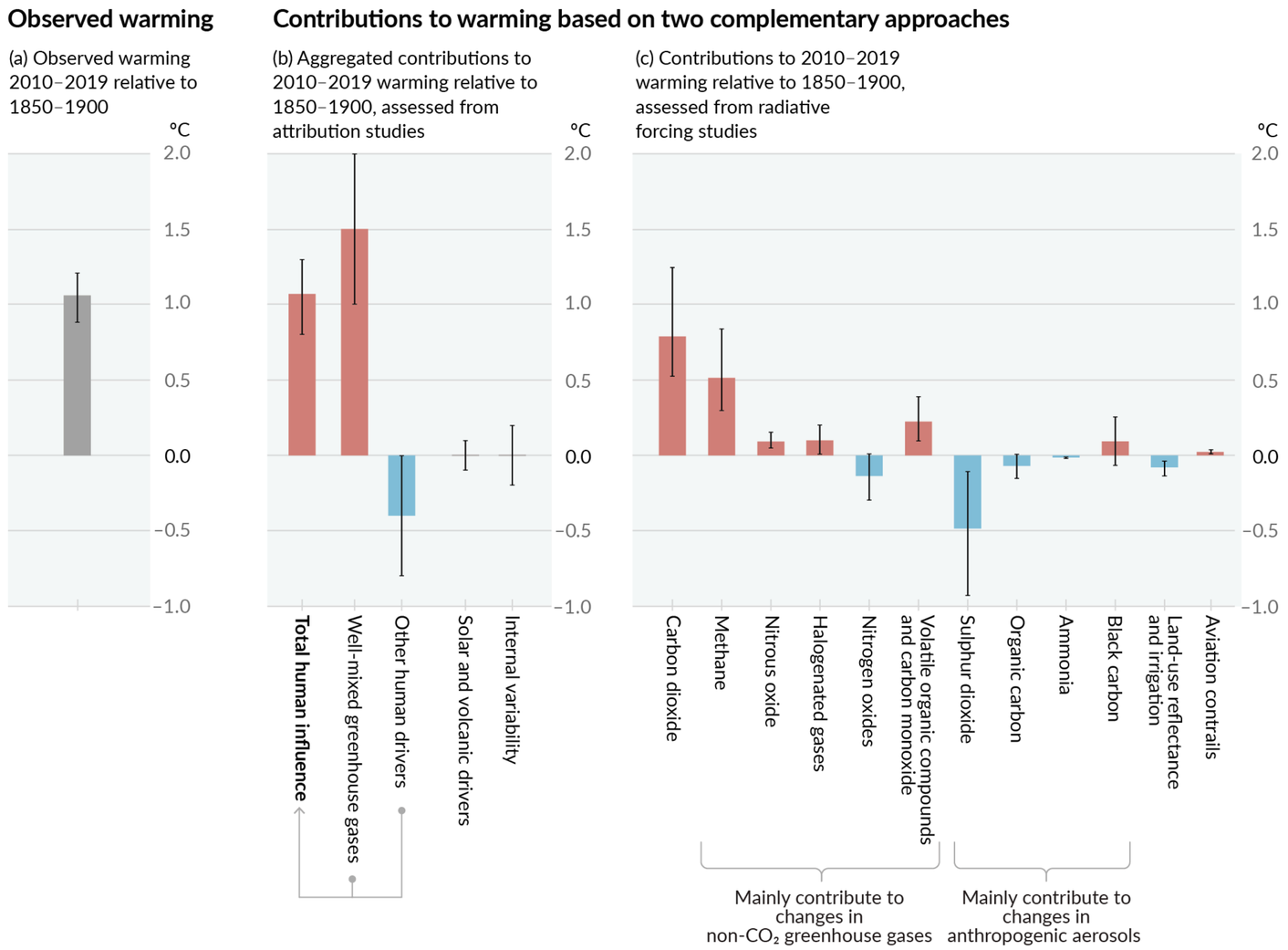
[FIFTH SERIES.]

APRIL 1896.

XXXI. *On the Influence of Carbonic Acid in the Air upon the Temperature of the Ground.* By Prof. SVANTE ARRHENIUS.

Svante Arrhenius (1859-1927)

Anthropogenic forcing of the climate system



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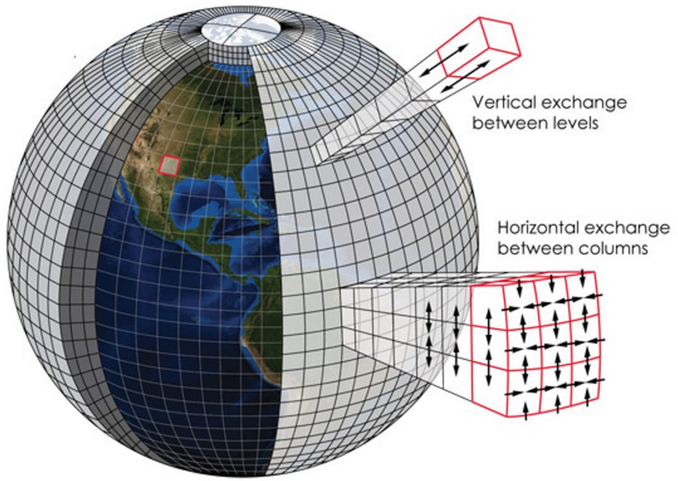
Climate models

Numerical models of the Earth System

Atmosphere and ocean physics



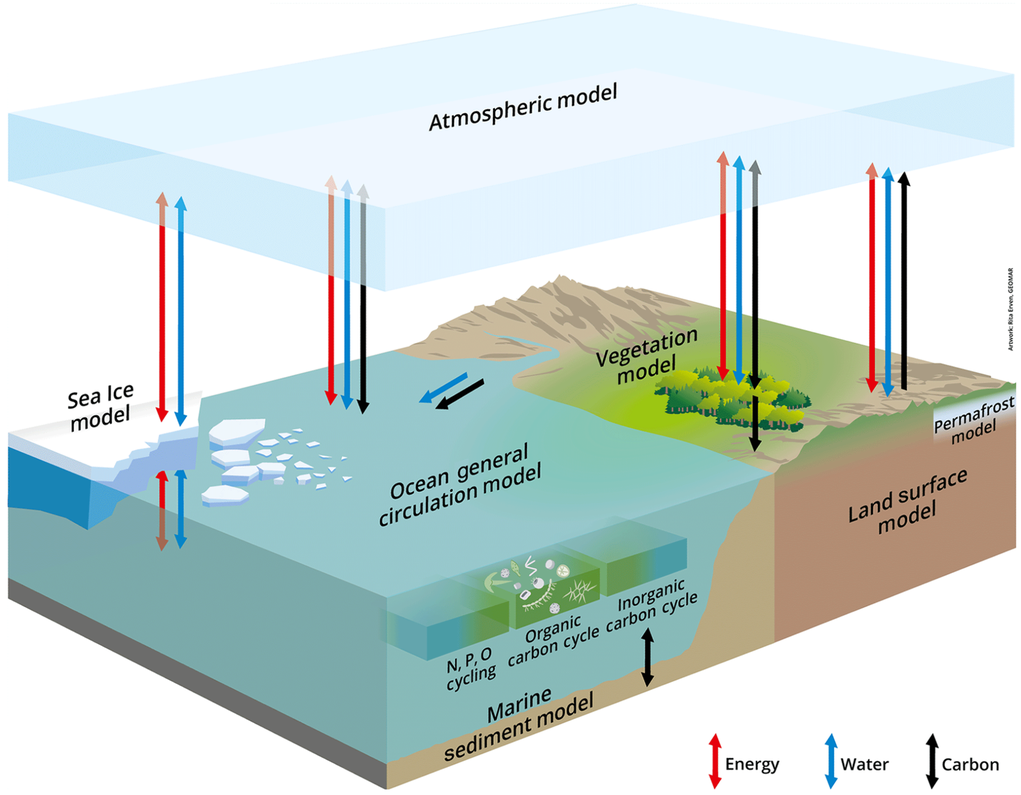
$$\frac{d\vec{v}}{dt} = -\frac{1}{\rho}\vec{\nabla}p - \vec{g} + \vec{F}_{fric} - 2\vec{\Omega}\times\vec{v}$$

Biological/chemical principles

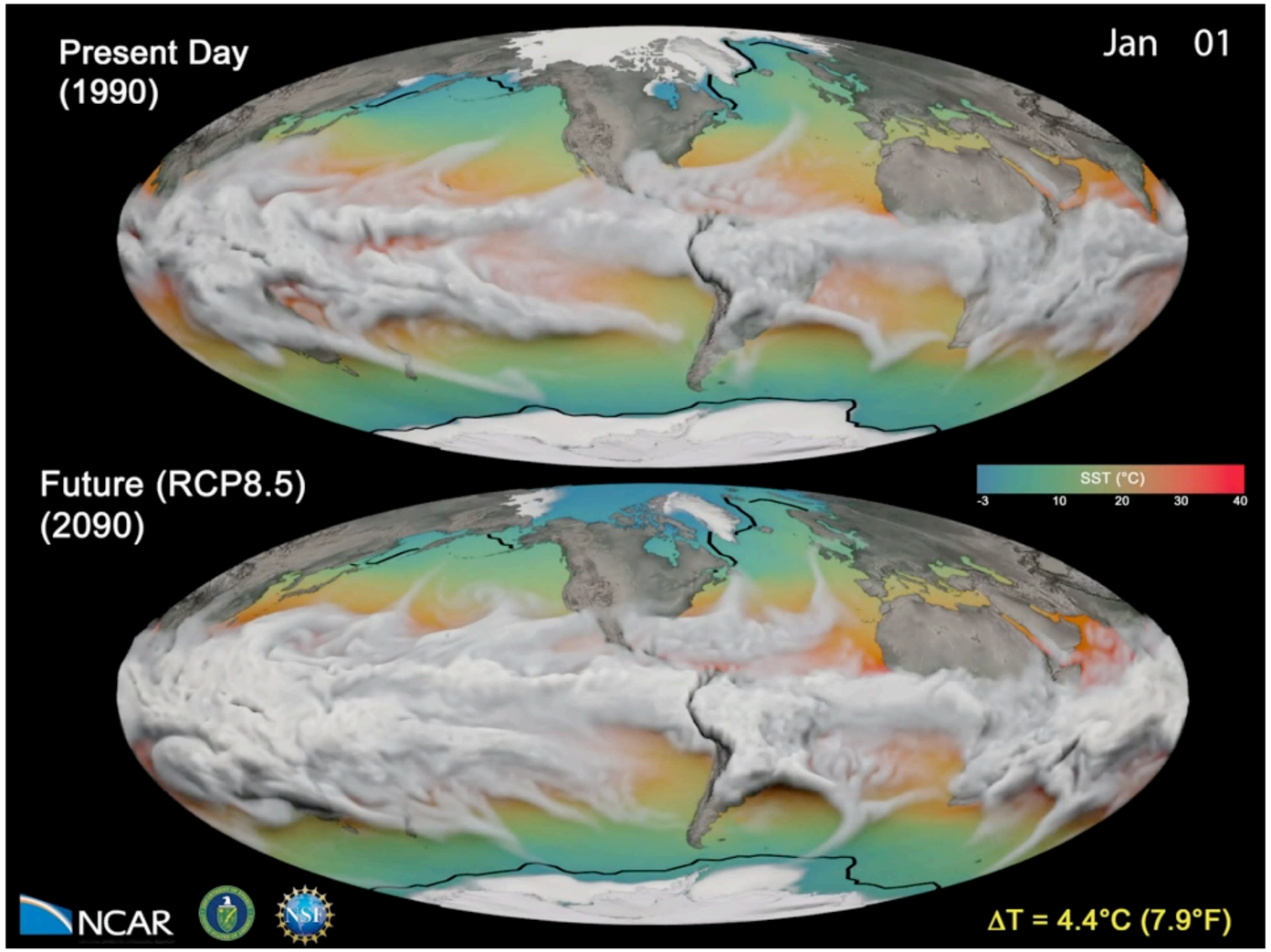
$$\vec{F} = M\vec{a}$$

e.g. momentum conservation

Atmosphere, ocean, land biogeochemistry

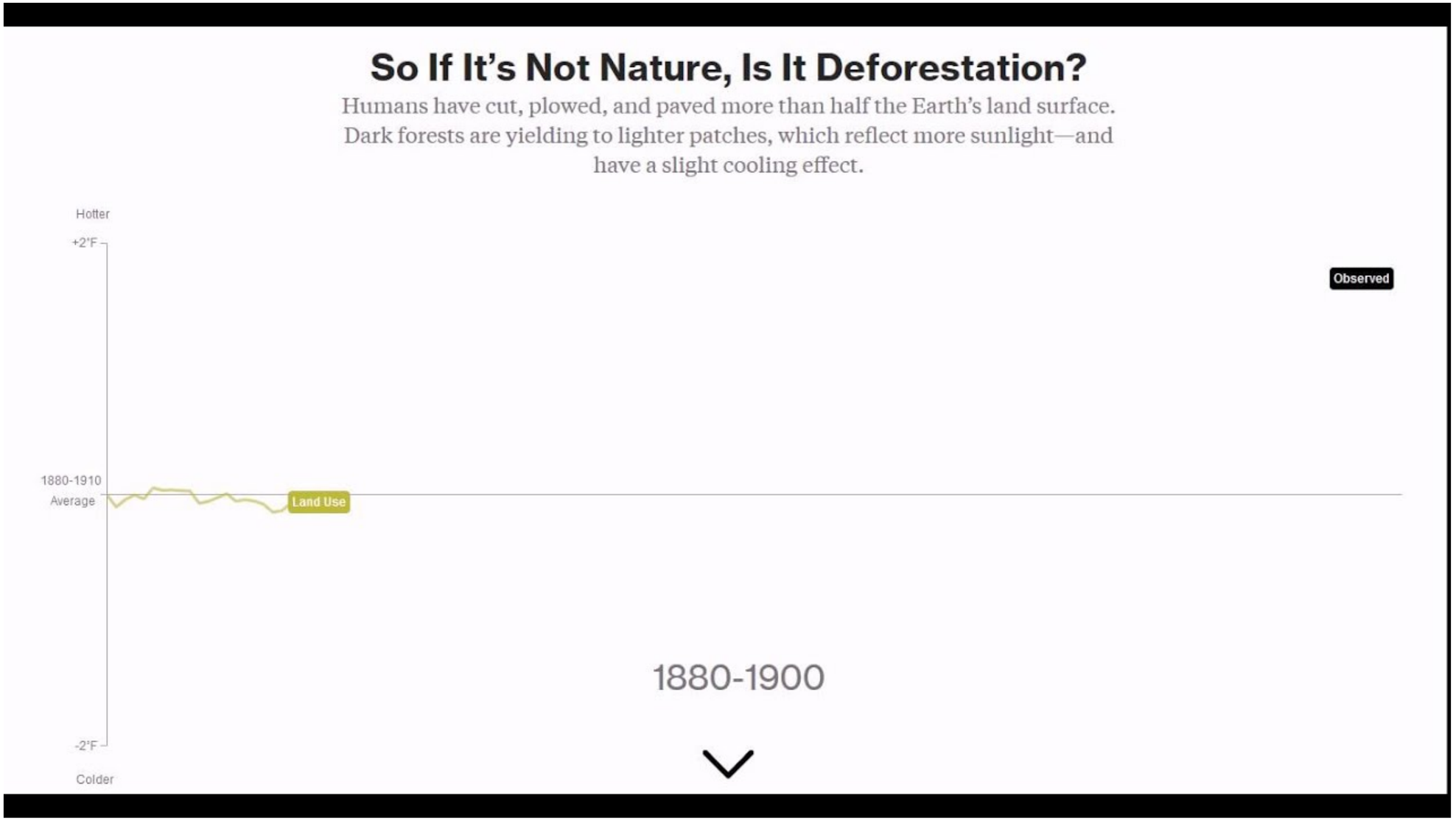


Numerical simulations of present and future climates



Community Earth System Model, NCAR

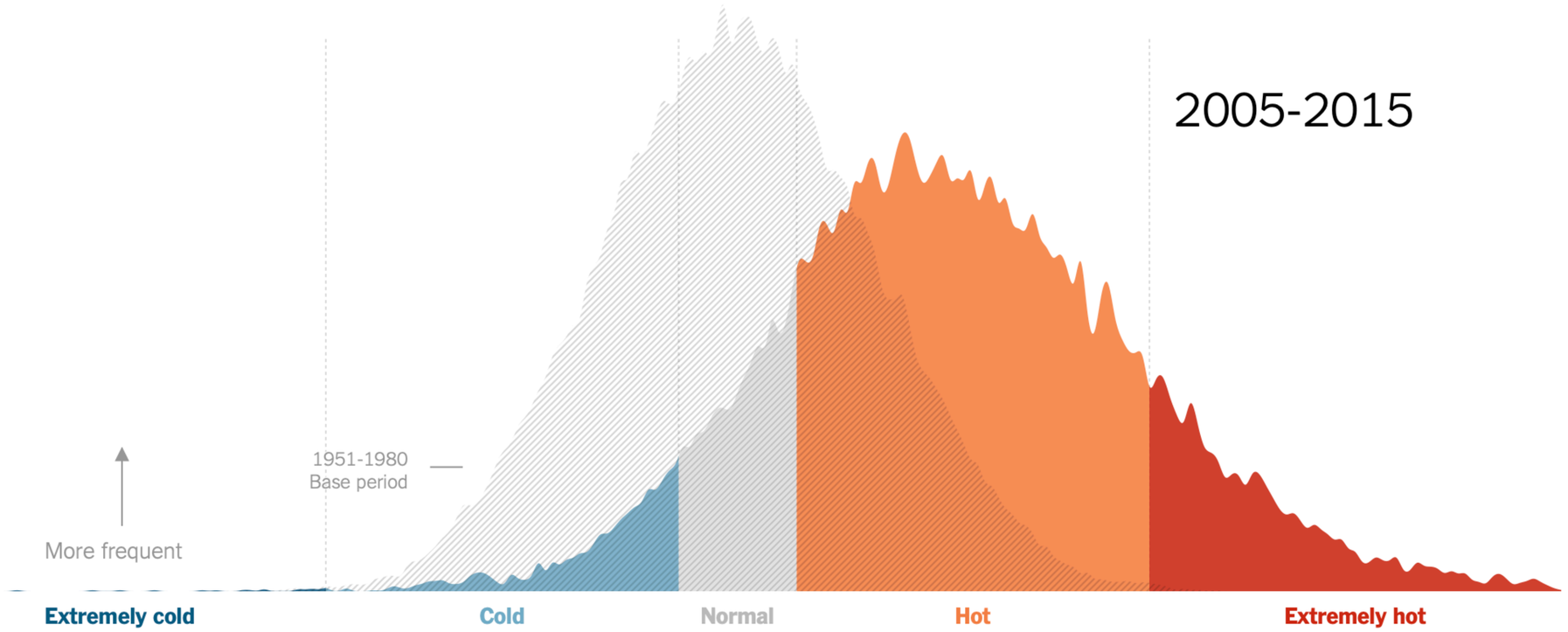
Climate forcing by natural versus human forcing



Consequences

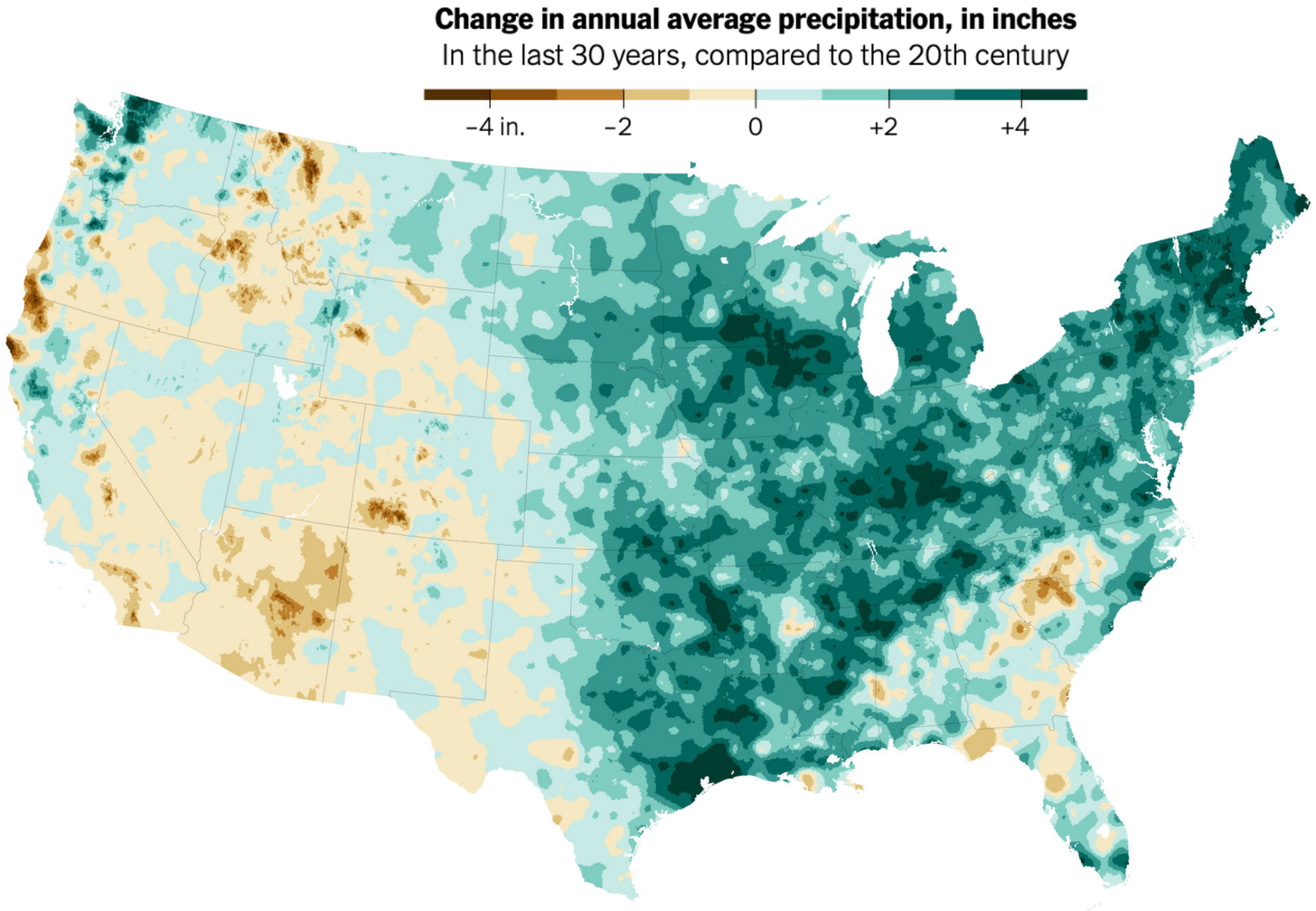
Extreme weather events

Summer temperature in the Northern Hemisphere



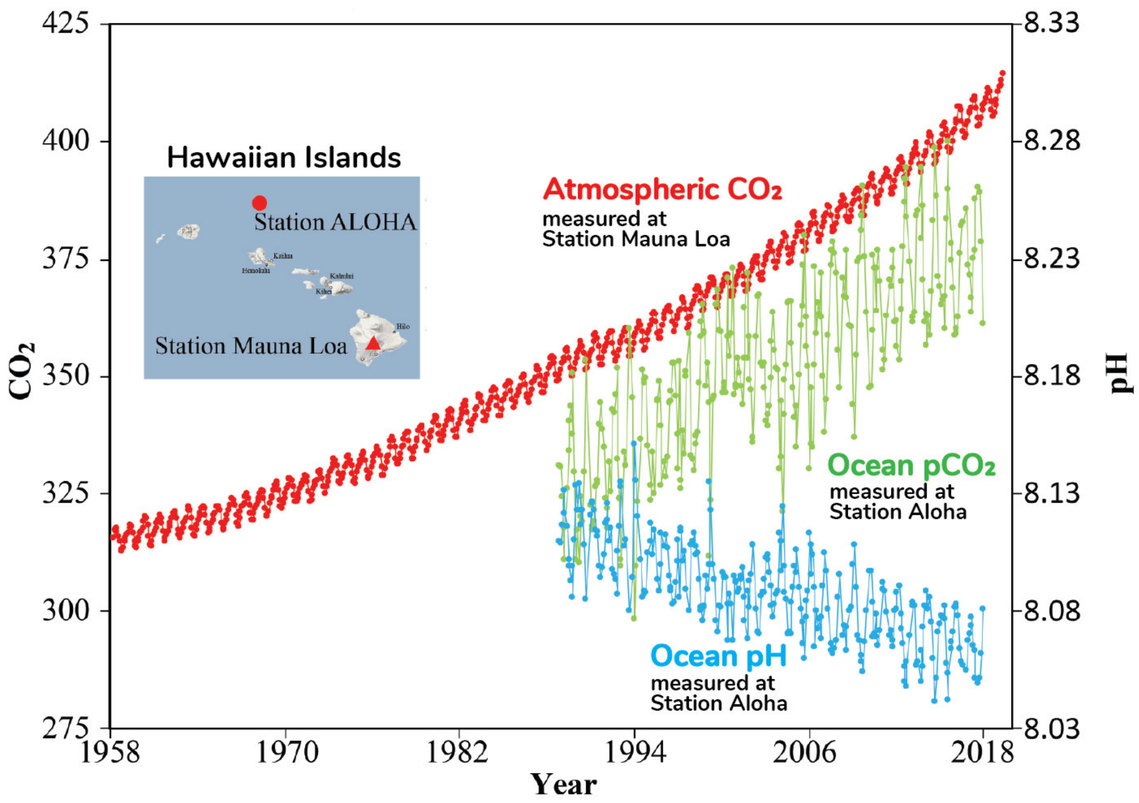
Sato and Hansen, modified by New York Times

Precipitation



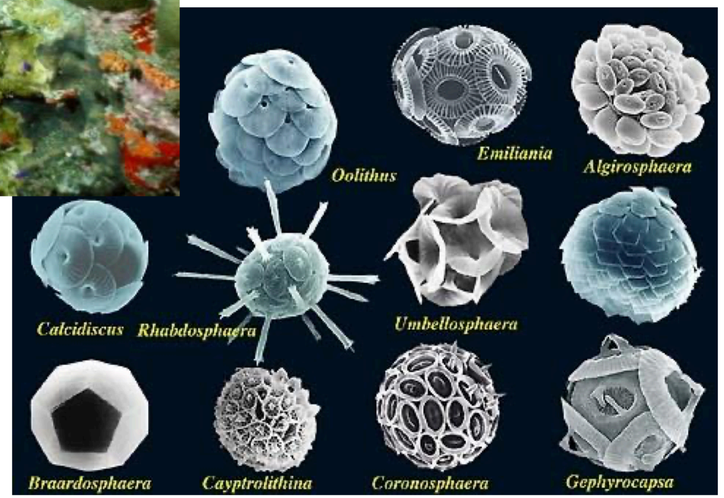
New York Times

Ocean acidification



- Acidification through CO_2 threatens marine life

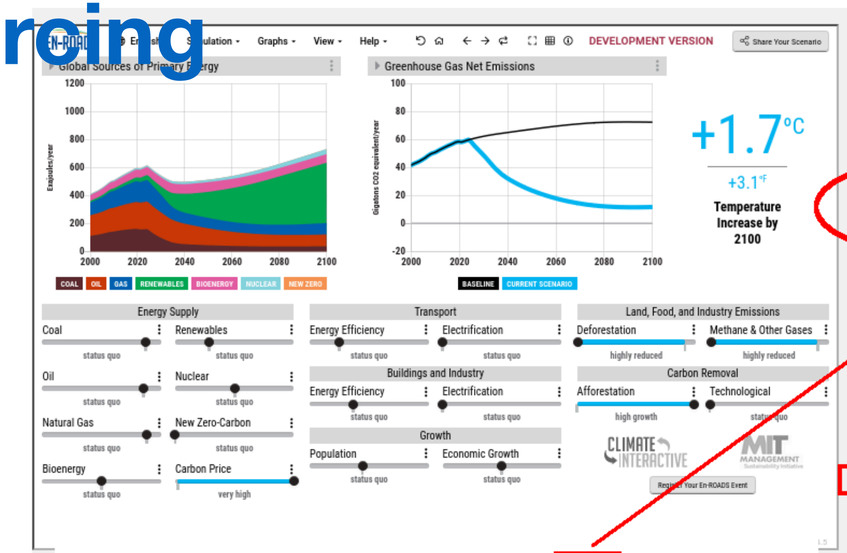
Plankton



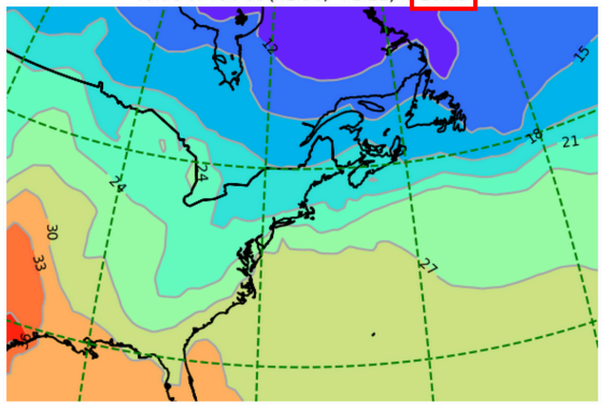
Sea level rise

NASA's Goddard Space Flight Center

Predicting climate response to anthropogenic forcing



value of T90 at (42.38, -71.11) is 20.69



Global mean temperature anomaly:
1.67
Click on the globe to set center position, or enter the lat/lon
Lat 42.38 Lon -71.11
Set lat/lon
Or select by country/city

Where

COUNTRY: United States
CITY: Cambridge
42.38, -71.11

Show: upper 90% in deg C
Month: Sep

Year: 2075

Display: Means New realization

Data:
global mean T anomaly = 1.67 °C
year = 2075 , month = 9
==========
latitude = 42.38
longitude = -71.11
units: degrees C
Temperature anomaly = 2.93
Temperature = 18.73
Standard deviation = 1.09
Difference from base case = -1.72

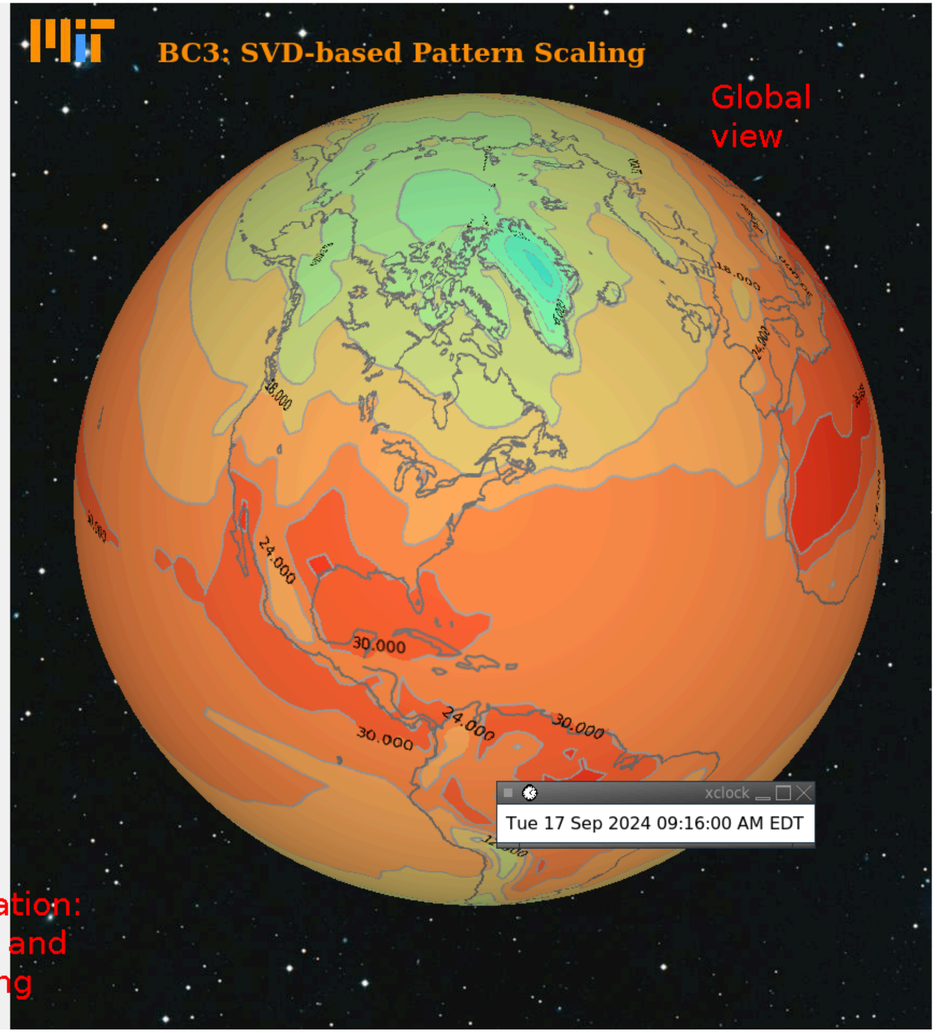
==========

Zoom level

Current: 1

Click on map to change center

More complete information: expected temperature and 10% probablilty of being higher than T90



BC3 group: Glenn Flierl , Shariqah Hossain (Gosha Geogdzhayev, Andre Souza)